

DRAFT

DETAIL SPECIFICATION SHEET

CABLES, CONNECTORS, ELECTRICAL, RUGGED, USB TYPE-C®, QUICK DISCONNECT
ENVIRONMENT RESISTANT HERMETICALLY SEALED WITH FIXED CONTACTS, GENERAL
SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of War.

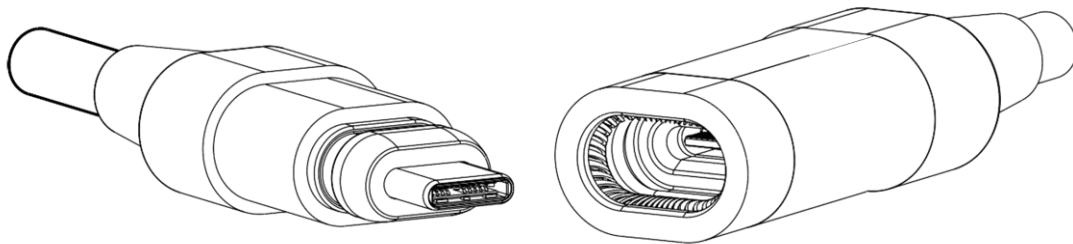


FIGURE 1: USB-CM CONNECTORS

CONTENTS

SECTIONS

1.	SCOPE.....	6
1.1.	Scope.....	6
2.	APPLICABLE DOCUMENTS.....	6
2.1.	General.....	6
2.2.	Government documents.....	6
2.3.	Non-Government publications.....	6
2.4.	Order of precedence.....	7
3.	REQUIREMENTS.....	7
3.1.	Specification sheets.....	7
3.2.	Qualification.....	7
3.3.	Materials.....	8
3.4.	Assembly Types.....	9
3.5.	Assembly Components.....	10
3.6.	Assembly Rating.....	10
3.7.	Part Numbering.....	11
3.8.	Configuration interface and features.....	11
3.9.	USB Type-C® cable & connector.....	12
3.10.	Interchangeability.....	13
3.11.	Temperature shock.....	13
3.12.	Reserved.....	13
3.13.	Coupling and uncoupling force.....	13
3.14.	Mating / Unmating cycles.....	13
3.15.	Insulation resistance.....	13
3.16.	Dielectric withstanding voltage.....	14
3.17.	Water Jets (IP X6).....	14
3.18.	Connector and cable flexing.....	14
3.19.	Low pressure (Altitude).....	14
3.20.	Vibration.....	14
3.21.	RESERVED.....	14
3.22.	Humidity.....	14
3.23.	EMI/EMC.....	14
3.24.	Ozone exposure.....	14
3.25.	Immersion.....	15
3.26.	Contamination by fluids.....	15
3.27.	High temperature exposure.....	15

3.28.	Low temperature exposure	15
3.29.	Ice resistance.....	15
3.30.	Rain.....	15
3.31.	Dust or fine sand.....	15
3.32.	Hydrolytic stability.	15
3.33.	Freezing rain.....	15
3.34.	Salt Fog.....	16
3.35.	Marking.....	16
3.36.	Workmanship.....	16
3.37.	Change effectivity.....	16
3.38.	Disposition of stock.....	16
4.	VERIFICATION	16
4.1.	Classification of inspection.....	16
4.2.	Inspection conditions.....	16
4.3.	Material Inspection.....	17
4.4.	Conformance inspection.....	17
4.5.	Qualification inspection.....	20
4.6.	Basic Operational Test (BOT).....	24
4.7.	RESERVED.....	25
4.8.	RESERVED.....	25
4.9.	USB Type-C® cable & connector.....	25
4.10.	Interchangeability.....	25
4.11.	Temperature shock.....	25
4.12.	RESERVED	25
4.13.	Coupling and uncoupling force.....	25
4.14.	Mating / Unmating cycles.....	25
4.15.	Insulation resistance (IR).....	25
4.16.	Dielectric withstanding voltage (DWV).....	26
4.17.	Water Jets (IP X6).....	26
4.18.	Connector and cable flexing.....	26
4.19.	Low Pressure (Altitude).....	26
4.20.	Vibration.....	26
4.21.	RESERVED.....	27
4.22.	Humidity.....	27
4.23.	EMI/EMC.....	27
4.24.	Ozone exposure.....	28
4.25.	Immersion.....	28

4.26.	Contamination by fluids	28
4.27.	High temperature exposure.	28
4.28.	Low temperature exposure.	28
4.29.	Ice resistance.....	29
4.30.	Rain.....	29
4.31.	Dust or fine sand.....	29
4.32.	Hydrolytic stability.	29
4.33.	Freezing rain.	29
4.34.	Salt Fog.....	29
4.35.	Post test examination.	29
5.	PACKAGING	30
5.1.	Packaging.	30
6.	NOTES	30
6.1.	Intended use.	30
6.2.	Application guidance and restrictions.	30
6.3.	Derating factors.	30
6.4.	Military unique statement.....	30
6.5.	Acquisition requirements.	30
6.6.	Qualification.	30
6.7.	Definitions.	30
6.8.	Subject term (key word) listing.....	31
6.9.	Guidance information.....	31
6.10.	Tin whisker growth.....	32
6.11.	Changes from previous issue.	32

FIGURES

Figure 1: USB-CM Connectors	1
Figure 2: USB-CM connector component references	8
Figure 3: MIL-DTL-32812 cable/connector family	9
Figure 4: Part Number designation	11
Figure 5: Interfacing Dimensions, Female	33
Figure 6: Interfacing Dimensions, Male	34
Figure 7: Plug, Receptacle, and Cable General Dimensions.	35
Figure 8: USB 2.0 Plug and Cable General Dimensions.	36
Figure 9: Right Angle Plug General Dimensions.	37
Figure 10: Locking Receptacle General Strategies.	38
Figure 11 Locking receptacle max dimensions.	39
Figure 12: Bulkhead Receptacle General Dimensions.	40
Figure 13: Labeling.	41

TABLES

Table 1: Group A inspection.	18
Table 2: Group B inspection.	18
Table 3: Sampling plan.	19
Table 4: sample groups.....	20
Table 5: Qualification and group C periodic tests.	24
Table 6: Basic Operational test	24
Table 7: CS118 Discharge table.	27

1. SCOPE

1.1. Scope.

This specification covers a series of environmentally resistant electrical cables and connectors based on the USB Type-C® specification, called USB-C Military (USB-CM). The connectors designated herein are intended to meet all the performance requirements of the USB Type-C® cables and connectors commercial specification and are enhanced to operate under extreme environmental and temperature conditions required for US Department of War use. Where a conflict between the Type-C® specification and this document exists, the specification in this document shall take precedence. See 6.1 for intended use, application notes, application restrictions, and derating factors.

2. APPLICABLE DOCUMENTS

2.1. General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1. Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.5).

DEPARTMENT OF WAR STANDARDS

MIL-STD-790	Standard Practice for Established Reliability and High Reliability Qualified Products List (QPL) Systems for Electrical, Electronic, and Fiber Optic Parts.
MIL-STD-461G	REQUIREMENTS FOR THE CONTROL OF ELECTROMAGNETIC INTERFERENCE CHARACTERISTICS OF SUBSYSTEMS AND EQUIPMENT
MIL-STD-810H w/Change 1	ENVIRONMENTAL ENGINEERING CONSIDERATIONS AND LABORATORY TESTS
MIL-STD-889	Dissimilar Metals.
MIL-STD-1285	Marking of Electrical and Electronic Parts.

(Copies of these documents are available online at <http://quicksearch.dla.mil>.)

2.2.2. Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

NONE.

2.3. Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.5).

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

EIA-364	Electrical Connector/Socket Test Procedures including Environmental Classifications.
---------	--

EIA-364-14 Ozone Exposure Test Procedure for Electrical Connectors.
EIA-364-32 Thermal Shock (Temperature Cycling) Test Procedure for Electrical Connectors and Sockets.
EIA-364-114 Coupling and Uncoupling Force Test Procedure for Electrical Connectors, Sockets and Applicable Accessories.

(Copies of these documents are available online at <http://eciaonline.org> or from the Electronic Components Industry Association (ECIA), 1111 Alderman Dr., Suite 400, Alpharetta, GA 30005-4175.)

ASME INTERNATIONAL

ASME Y14.5 Dimensioning and Tolerancing.

(Copies of this document are available online at <https://www.asme.org> or from the ASME International, Three Park Avenue, New York, NY 10016-5990.)

ASTM INTERNATIONAL

ASTM B545 Standard Specification for Electrodeposited Coatings of Tin.

ASTM D570 Standard Test Method for Water Absorption of Plastics.

(Copies of these documents are available at <https://www.astm.org> or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, Pennsylvania 19428-2959.)

SAE INTERNATIONAL

SAE-AIR1351 Manufacturers' Identification of Aerospace Electrical and Electronic Wiring Devices and Accessories.

SAE-EIA557 Statistical Process Control Systems.

(Copies of these documents are available at <https://www.sae.org> or SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 (inside USA and Canada) or 724-776-4970 (outside USA), or email: CustomerService@sae.org.)

USB-IF

USB Type-C® r2.4 Universal Serial Bus Type-C® Cable and Connector Specification

USB PD r3.2 v1.1 Universal Serial Bus Power Delivery Specification

(Copies of these documents are available at <https://www.usb.org/>)

2.4. Order of precedence.

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. Specification sheets.

The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2. Qualification.

Connectors and accessories furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.5 and 6.6). After qualification, if the manufacturer seeks to make design changes or process changes to qualified product, or if the manufacturer seeks to use alternate / substitute materials, then the manufacturer shall coordinate with the qualifying activity and obtain its approval for proposed changes prior to implementing design or process changes to qualified product.

3.2.1. Qualified Products List (QPL) system.

The manufacturer shall establish and maintain a quality system that allows its parts that are covered by this specification to be listed on the QPL. Requirements for this system are specified in MIL-STD-790. In addition, the manufacturer shall establish a Statistical Process Control (SPC) system that meets the requirements specified in SAE-EIA557.

3.3. Materials.

3.3.1. Critical interface materials.

Materials shall be as specified herein. When a definite material is not specified, a material shall be used which shall enable the connector to meet the performance requirements of this specification.

3.3.1.1. Metals.

Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion.

3.3.1.2. Dissimilar metals and compatible couples.

When dissimilar metals are used in intimate contact with each other, protection against galvanic corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active galvanic corrosion (particularly brass, copper, or steel used in contact with aluminum or an aluminum alloy) is not acceptable. However, metal plating of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The use of dissimilar metals separated by a suitable insulating material is also permitted. Dissimilar metals and compatible couples are specified in MIL-STD-889.

3.3.1.3. Pure tin.

The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of connectors, their components and solder shall not exceed 97 percent, by mass. Tin may be alloyed with a minimum of 3 percent lead, by mass (see 6.10).

3.3.2. Components.

Materials for specific components of the connector shall be as follows:

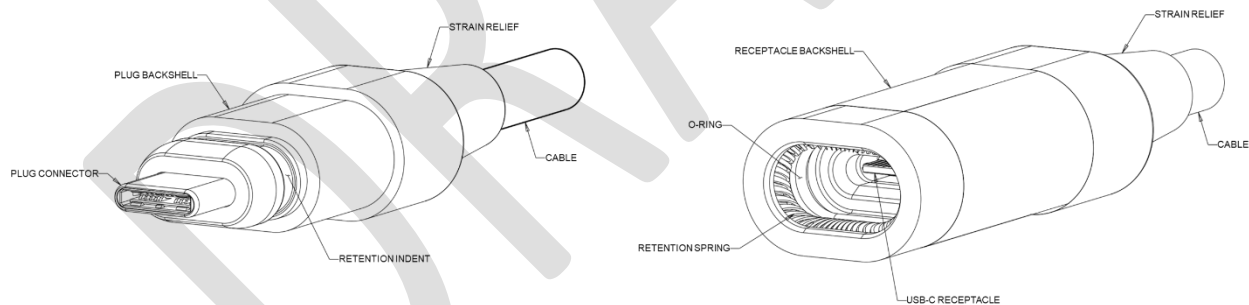


FIGURE 2: USB-CM connector component references

- a. Cable –
 - a. Conductor - copper with tin or silver plating.
 - b. SDP Shield - AL foil or AL/mylar foil.
 - c. Coaxial shield - copper strand.
 - d. Braid - tin plated copper or aluminum.
 - e. Jacket - PVC or halogen free substitute material.
- b. Strain Relief - Thermoplastic elastomer (TPE) or equivalent
- c. Backshell - Coated aluminum or equivalent non-ferrous material
- d. Connector shells - Corrosion-resistant steel.
- e. Retention Spring - Heat-treated beryllium copper or corrosion-resistant steel.
- f. Potting compound – epoxy adhesive

g. Gaskets, grommet, and interface seals - Silicone or fluorosilicone elastomer.

3.3.3. Fungus resistance.

Materials used in the construction of these connectors shall be fungus inert (see 4.3.1).

3.3.4. Recycled, recovered, or environmentally preferable, or biobased materials.

Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.5. Hazardous materials.

The use of environmentally preferable materials in the construction of these connectors is encouraged. See 6.9 for guidance.

3.4. Assembly Types.

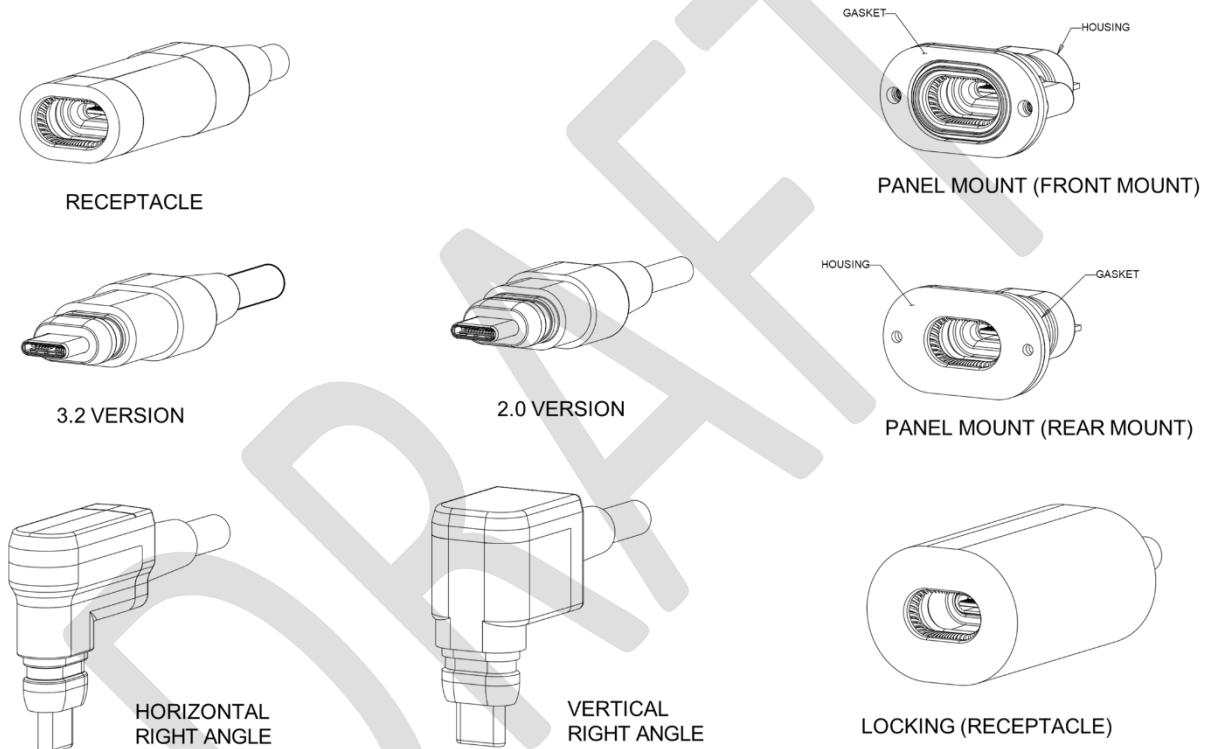


FIGURE 3. MIL-DTL-32812 cable/connector family.

3.4.1. Cable – Plug-to-Plug.

This assembly consists of a single length of cable with a plug at each of the two ends.

3.4.2. Cable – Plug-to-Receptacle.

This assembly consists of a single length of cable with a plug at the first end and a receptacle at the second end.

3.4.3. Cable – Receptacle-to-Receptacle.

Assemblies of this type are disallowed.

3.4.4. Connector – Bulkhead.

This class includes panel mount receptacles. Only one connector is specified. There is no cable associated with this assembly type.

3.5. Assembly Components.3.5.1. Plug.

All plug components shall share common interfacing dimensions and be compatible with breakaway or locking receptacles.

3.5.1.1. Straight Plug – 2.0 Version.

Dimensions shall be as shown in Figure 8.

3.5.1.2. Straight Plug – 3.2 Version.

Dimensions shall be as shown in Figure 7.

3.5.1.3. Horizontal Right-Angle Plug.

Dimensions shall be as shown in Figure 9.

3.5.1.4. Vertical Right-Angle Plug.

RESERVED.

3.5.2. Receptacle.3.5.2.1. Breakaway Receptacle.

Dimensions shall be as shown in Figure 5. The spring material and dimensions shall be selected to provide between 14 ± 3 lbs. uncoupling force for a minimum of 500 mating/unmating cycles.

3.5.2.2. Locking Receptacle.

Dimensions shall be as shown in Figure 11. The sliding retention collar locking female connectors shall provide a minimum uncoupling force of 45 lbs. The sliding retention collar actuation force shall be between 4 and 6 lbs.

3.5.2.3. Bulkhead Receptacle.

Dimensions shall be as shown in Figure 12.

3.5.3. Cable.

Applicable to Plug to Plug and Plug to Receptacle cable assemblies. Cable lengths shall be specified from the end of the strain relief (ex. Figure 7).

3.6. Assembly Rating.

Assemblies shall be assigned and adhere to USB power and data rates. See 3.35 for marking information.

3.6.1. Power Rating.

Assemblies shall be identified with a USB-PD power rating (see 3.9.1).

3.6.2. Data Rate.

Assemblies shall be identified with a USB data rate specification (see 3.9.1).

3.7. Part Numbering.

Part Number		32812-	060	20HS	SP	SP	24
Designator	32812- = USB-CM Product						
USB Power	060 = 60 Watt 100 = 100 Watt 240 = 240 Watt						
USB Data Rate	10LS = USB 1.0/Low-Speed 11FS = USB 1.1/Full-Speed 20HS = USB 2.0/Hi-Speed 3211 = USB 3.2 Gen 1x1 5 Gbps 3221 = USB 3.2 Gen 2x1 10 Gbps 3212 = USB 3.2 Gen 1x2 10 Gbps 3222 = USB 3.2 Gen 2x2 20 Gbps 4022 = USB 4.0 Gen 2x2 20 Gbps 4032 = USB 4.0 Gen 3x2 40 Gbps						
Termination 1	SP = Straight Plug HP = Horizontal Right-Angle Plug VP = Vertical Right-Angle Plug BR = Breakaway Receptacle LR = Locking Receptacle BF = Bulkhead Receptacle, Front Panel Mount / 1 BR = Bulkhead Receptacle, Rear Panel Mount / 1						
Termination 2	SP = Straight Plug HP = Horizontal Right-Angle Plug VP = Vertical Right-Angle Plug BR = Breakaway Receptacle LR = Locking Receptacle BF = Bulkhead Receptacle, Front Panel Mount / 1 BR = Bulkhead Receptacle, Rear Panel Mount / 1						
Length	Cable Length, rounded to nearest whole inch / 2						

Notes:

/1 Bulkhead Receptacle is repeated as termination 2 for this type i.e. "BFBF".

/2 Use "00" for Bulkhead Receptacle.

FIGURE 4: PART NUMBER DESIGNATION3.8. Configuration interface and features.

Connectors and accessories shall be configured to withstand normal handling during installation and maintenance while in service. All accessories designed to be used with MIL-DTL-32812 connectors shall conform to these dimensions. Dimensioning and tolerancing shall be in accordance with ASME Y14.5. Unless otherwise specified, dimensions apply after plating.

3.8.1. Contacts.

Contacts shall be fixed, as specified.

3.8.1.1. Dust caps/containers.

Dust caps shall be placed on the mating end of each connector for shipping. In lieu of dust caps on both ends, receptacles may be packaged in rigid or semi-rigid containers designed to prevent damage to the end of the connector.

3.8.2. Coupling.

Connectors shall be coupled to counterpart connectors by the following coupling methods:

Plugs shall use a common set of dimensions and materials for their mating interface and shall be interchangeable with one another.

Breakaway receptacles shall use a common set of dimensions and materials for their mating interface and shall be interchangeable with one another and compatible with plugs.

Locking receptacles shall use a common set of dimensions and materials for their mating interface and shall be interchangeable with one another and compatible with plugs.

All plugs shall be mateable and interoperable with both breakaway or locking receptacles.

3.8.2.1. Ease of coupling.

Counterpart connectors of any arrangement shall be capable of being fully coupled and uncoupled in a normal and accessible location without the use of tools.

Regardless of retention method, the design shall facilitate mating and unmating while wearing gloves. Specifically, the connector overmold tail and cable should be of sufficient size and rigidity to grasp and manipulate the retention feature while wearing gloves.

3.8.2.2. Breakaway connectors.

Breakaway receptacles shall use an internal retention spring or equivalent retention mechanism. This mechanism is intended to interface with the retention indent feature. Breakaway connectors shall disconnect without the use of tools.

3.8.2.3. Locking connectors.

Multiple approaches to locking receptacles are allowed. Locking receptacles may utilize a techniques shown in Figure 10. This mechanism is intended to interface with the retention indent feature. Dimensions of this mechanism shall be specified as shown in Figure 11.

3.8.2.4. Polarization of connector shells.

Plugs and receptacles shall be mateable in a bidirectional manner. Connectors shall not rely on any keying mechanism to establish orientation.

3.8.2.5. Lubrication.

Coupling ramps, retention springs, gaskets, and locking springs of connectors may be coated during the manufacturing process with a suitable lubricant. Lubricants shall be non-flaking and galvanically compatible. Lubricants shall not migrate into contact areas.

3.9. USB Type-C® cable & connector.

3.9.1. Applicable specifications.

All requirements of the Universal Serial Bus Type-C® Cable and Connector Specification, Release 2.4, October 2024 (Hereafter, "Type-C® specification") are applicable to cables and connectors defined by this specification. Where a conflict between the Type-C® specification and this document exists, the specification in this document shall take precedence.

3.9.1.1. USB 3.2 GEN 2x2, 240W.

Cable/connector assemblies of this type shall utilize USB Full-Featured Type-C® receptacles for USB 3.2 platforms.

Cable/connector assemblies of this type shall utilize USB Full-Featured Type-C® plugs.

Cable/connector assemblies of this type shall utilize USB Full-Featured Type-C® cable.

All cable/connector assemblies of this type shall be electronically marked.

All cables/connector assemblies of this type shall be physically marked with data/power capabilities. Utilize USB-IF markings for power and data rates on the backshell where possible. Additional information should be on a heat-shrink wire information band.

3.9.1.2. USB 2.0, 60W.

Cable/connector assemblies of this type shall utilize USB 2.0 Type-C® plugs and may utilize USB Full-Featured Type-C® plugs. Only the contacts necessary to support USB PD and USB 2.0 are required in the plug.

Cable/connector assemblies of this type may be electronically marked.

All cables/connector assemblies of this type shall be physically marked with data/power capabilities. Utilize USB-IF markings for power and data rates on the backshell where possible. Additional information should be on a heat-shrink wire information band.

3.9.1.3. Vconn

Cable/connector assemblies shall include the optional Vconn if Fully Featured. Cable/connector assemblies may include the optional Vconn if USB 2.0.

3.10. Interchangeability.

All connectors having the same Part or Identifying Number (PIN) shall be completely interchangeable with each other with respect to installation and performance. Verify in accordance with 4.10.

3.11. Temperature shock.

When tested as specified in 4.11, there shall be no damage detrimental to the operation of the connector.

3.12. Reserved

3.13. Coupling and uncoupling force.

See 3.5.2. When tested as specified in 4.13, the coupling force for mating and the uncoupling force for unmating of counterpart plugs and receptacles shall apply to mating of connectors. Coupling and uncoupling force requirements do not apply to mating plugs to and from dummy stowage receptacles and/or covers.

3.14. Mating / Unmating cycles.

The durability rating shall be 10,000 cycles minimum. The durability test shall be done at a rate of 500 ± 50 cycles per hour. When tested as specified in 4.14, the connectors shall show no defects detrimental to the operation of the connectors and shall meet the subsequent test requirements (see 4.5.3).

3.15. Insulation resistance.

3.15.1. Insulation resistance at ambient temperature.

When tested as specified in 4.15.1, the insulation resistance between any pair of contacts and between any contact and the shell shall be greater than 100 MΩ. Insulation resistance after altitude immersion shall be 100 MΩ minimum. Insulation resistance after humidity shall be 100 MΩ minimum.

3.15.2. Insulation resistance at elevated temperature.

When tested as specified in 4.15.2, the insulation resistance between any pair of contacts and between any contact and the shell shall be greater than 100 MΩ.

3.16. Dielectric withstanding voltage.

When tested as specified in 4.16.1 or 4.16.2, the maximum leakage current shall be 2 milliamperes, and there shall be no evidence of electric breakdown or flashover.

3.17. Water Jets (IP X6).

Unmated connectors shall meet the ingress protection IP X6 standard. When tested as specified in 4.17, unmated connectors shall show no lifting of plating or exposure of basis material under three power (3X) magnification which adversely affects performance.

3.18. Connector and cable flexing.

When tested as specified in 4.18, connectors shall show no evidence of damage detrimental to their normal operation, nor shall there be any interruption of electrical and/or data continuity.

3.19. Low pressure (Altitude).

When tested as specified in 4.19, the connectors shall meet the requirements of dielectric withstanding voltage at sea level specified in 3.16 and insulation resistance at ambient temperature specified in 3.15.1.

3.20. Vibration.

When tested as specified in 4.20, there shall be no electrical discontinuity and there shall be no disengagement of the mated connectors, backing off of the coupling mechanism, evidence of cracking, breaking, or loosening of parts.

3.21. RESERVED.3.22. Humidity.

When tested as specified in 4.22, mated and unmated connectors shall show no deterioration which will adversely affect performance of the connector.

3.23. EMI/EMC.

Cables and connectors shall comply with the following emission and susceptibility requirements of MIL-STD-461.

3.23.1. CS118 (Conducted Susceptibility, Personnel Borne Electrostatic Discharge).

Cables and connectors shall withstand (survive) ESD to $\pm 8\text{kV}$ contact discharge and $\pm 15\text{kV}$ air discharge on any surface, including exposed pins, in both operating and non-operating mode. Verify in accordance with 4.23.1.

3.23.2. RE102 (Radiated Emissions, Electric Field).

Cables and connectors, when operating at their maximum rated data transfer rate and power delivery capacity, shall control radiated electric field emissions. The emissions shall not exceed the "Ground, Army" limits specified in MIL-STD-461, Method RE102, over the frequency range of 2 MHz to 18 GHz. Verify in accordance with 4.23.3.

3.23.3. RS103 (Radiated Susceptibility, Electric Field).

Cables and connectors shall operate without any sign of susceptibility, performance degradation, or deviation from specified functionality when subjected to radiated electric fields over the frequency range of 2 MHz to 2.7 GHz. Verify in accordance with 4.23.3.

3.24. Ozone exposure.

When tested as specified in 4.24, cables and connectors shall show no evidence of cracking of dielectric material or other damage due to ozone exposure that will adversely affect performance.

3.25. Immersion.

Applicable to initial qualification only. When tested as specified in 4.25, connectors shall meet the requirements for dielectric withstanding voltage (see 3.16). Mated and unmated connectors shall demonstrate immersion in fresh water and salt water to a specific profile.

3.25.1. Combat swimmer.

Mated and unmated connectors shall withstand depth of 33ft $\pm$ 1ft for 10 hours $\pm$ 0.25 hour.

3.25.2. RESERVED3.26. Contamination by fluids
RESERVED3.27. High temperature exposure.

When tested as specified in 4.27, connectors shall perform satisfactorily and pass succeeding tests in the qualification test sequence.

3.27.1. High temperature storage.

Mated and unmated connectors shall withstand storage temperatures of 160°F.

3.27.2. High temperature operation.

Mated and unmated connectors shall withstand operational temperatures of 130°F.

3.28. Low temperature exposure.

When tested as specified in 4.28, connectors shall perform satisfactorily and pass succeeding tests in the qualification test sequence.

3.28.1. Low temperature storage.

Mated and unmated connectors shall withstand storage temperatures of -40°F.

3.28.2. Low temperature operation.

Mated and unmated connectors shall withstand operational temperatures of -40°F.

3.29. Ice resistance.

When tested as specified in 4.29, connectors shall pass succeeding tests in the qualification table. When tested in accordance with 4.13, uncoupling and recoupling force shall not exceed the values specified in 3.13 by more than 25 percent.

3.30. Rain.

When tested as specified in 4.30, unmated connectors shall exhibit no detrimental effects following exposure. Mated pairs shall be capable of operating while exposed without degradation in performance or damage to the connectors or cables.

3.31. Dust or fine sand.

When tested as specified in 4.31, connectors shall pass succeeding tests in the qualification table. When tested as specified in 4.13, uncoupling and recoupling force shall not exceed the values specified in 3.13 by more than 25 percent.

3.32. Hydrolytic stability.

When tested as specified in 4.32, the connector shall be without defects detrimental to mechanical performance. There shall be no increase in the connector weight greater than 0.75 percent.

3.33. Freezing rain.

When tested as specified in 4.33, connectors shall pass succeeding tests in the qualification table. When tested in accordance with 4.13, uncoupling and recoupling force shall not exceed the values specified in 3.13 by more than 25 percent.

3.34. Salt Fog.

When tested as specified in 4.34, unmated connectors shall exhibit no detrimental effects following exposure. Mated pairs shall be capable of operating while exposed without degradation in performance or damage to the connectors or cables.

3.35. Marking.

Cable labeling shall be as specified in Figure 13. Cable wrap shall be marked with the appropriate Part Number (see 3.7) at each end.

3.36. Workmanship.

The connector shall be fabricated in a manner such that the criteria for appearance, fit and adherence to specified tolerances are observed. Particular attention shall be given to neatness and thoroughness of marking parts, plating, welding, soldering, riveting, staking, and bonding. The visual examination under three power (3X) magnification shall include examination of each retention cavity for molding discrepancies, retention features, and excessive adhesive. The connectors shall be free from crazing, cracks, voids, pimples, chips, blisters, pinholes, sharp cutting edges, burrs, and other defects that will adversely affect life, serviceability, or appearance.

3.37. Change effectivity.

Unless otherwise specified by the preparing activity and/or the qualifying activity, all changes from the previous revision of MIL-DTL-32812 shall become effective within 90 days of the date of publication of the latest revision. If unable to implement changes within the 90-day time period, additional time shall be requested from the qualifying activity. Manufacturers that are QPL-listed and have concerns regarding possible changes to retention reporting requirements should contact the qualifying activity for clarification.

3.38. Disposition of stock.

If connectors and accessories produced to the previous revision continue to meet the requirements of this specification, then products manufactured in accordance with the previous revision may continue to be shipped from stock for an indefinite period. If the qualified products meet the requirements of the previous revision, but do not meet the requirements of the current revision, then the manufacturer and their selling agents or distributors have a period of 18 months from the date of the latest revision to purge or ship all inventory, unless otherwise coordinated with, and authorized by, the qualifying activity. Revision changes to connector marking requirements, such as changes and / or additions of tracer markings to the grommet or connector mating face, do not impact form, fit, or function. Products produced in accordance with previous revisions with the above-referenced types of changes that do not impact form, fit, or function, may be sold until stocks are depleted.

4. VERIFICATION

4.1. Classification of inspection.

The examination and testing of connectors shall be classified as follows:

4.1.1. Conformance inspection (see 4.4).4.1.2. Qualification inspection (see 4.5).4.2. Inspection conditions.

Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of EIA-364.

4.2.1. Test equipment and inspection facilities.

Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment (i.e., industry standard, military standard) shall be required.

4.2.1.1. QPL system.

The manufacturer shall establish and maintain a system to meet the requirements of MIL-STD-790 and the requirements herein (see 3.2.1). Evidence of such compliance shall be verified by the qualifying activity as a prerequisite for qualification and retention of qualification. Assembly plants shall be approved by the qualifying activity and listed on the applicable Qualified Products List. The qualified connector manufacturer shall certify that the assembly plant is approved for the distribution of the manufacturer's parts. Assembly plants shall use only piece parts (i.e. subcomponents, such as connector shells and inserts) produced by the qualified manufacturer. No incoming inspection other than visual examination is required of certified piece parts obtained from the qualified manufacturer, except when there is cause for rejection. Assemblies produced at the assembly plant shall be subjected to inspection to ensure that the assembly process conforms with that established at the qualified manufacturing plant. Quality control requirements, including Government inspection surveillance, shall be the same as required for the qualified connector manufacturer.

4.3. Material Inspection

Material inspection performed through certification (see 4.4)

4.3.1. Fungus resistance certification.

Certification to method 508 of MIL-STD-810 and MIL-HDBK-454 Guideline 4 is required (see 3.3.3). Testing is not required for materials listed as fungus inert.

4.4. Conformance inspection.4.4.1. Inspection of product for delivery.

Inspection of product for delivery shall consist of groups A and B inspections. The documentation requirements of the applicable military or industry test procedures do not apply.

4.4.1.1. Group A inspection.

Group A inspection shall consist of the inspections specified in Table 1.

4.4.1.1.1. Visual examination (group A inspection).

Each connector and accessory shall be visually examined for completeness, workmanship, and identification requirements. For those assemblies that require a gasket, the condition of the gasket shall be visually examined. Gaskets missing, twisted, buckled, kinked, or damaged in any way shall be cause for rejection.

4.4.1.1.2. Critical examinations (group A inspection).

The following dimensions and features shown on the pertinent military standard are considered critical.

- a. Mating diameters H, W, and R.
- b. Cable lengths.
- c. Contact pattern relative to datum features.
- d. Marking features (see 3.35).
- e. (locking receptacles) Retention collar operation.

Inspection	Requirement paragraph	Test paragraph
Visual and mechanical examination. <u>1/ 2/</u>	3.1, 3.3, 3.4, 3.10, 3.35 and 3.36	0

Critical examination. <u>1/</u> <u>2/</u>	3.1	4.4.1.1.2
Insulation resistance (IR). <u>1/</u> <u>2/</u> <u>3/</u>	3.15	4.15
Dielectric withstanding voltage (DWV). <u>1/</u> <u>2/</u> <u>3/</u>	3.16	4.16

1/ 100 percent inspection.

2/ The contractor may use in-process controls for this requirement.

3/ Test 100 percent of contacts. Points of measurement and application of voltage shall be between circuits consisting of each contact and its closest adjacent contact and between connector shell and closest contacts adjacent to the shell. Multiple circuits may be tested concurrently.

TABLE 1: Group A inspection.

4.4.1.2. Group B inspection.

Group B inspection shall consist of the inspections specified in Table 2 in the order shown and shall be made on sample units which have been subjected to and have passed the group A inspection.

Inspection	Requirement paragraph	Test paragraph	Number of samples
Coupling and uncoupling force. <u>1/</u> <u>2/</u>	3.13	4.13	As specified in Table 3
Connector and cable flexing. <u>1/</u> <u>2/</u> <u>3/</u>	3.18	4.18	
Post test examination. <u>1/</u> <u>2/</u> <u>3/</u>	3.35 and 3.36	4.35	

1/ The contractor may use in-process controls for this requirement.

2/ No failures permitted.

3/ Test 5 pieces.

TABLE 2: Group B inspection.

4.4.1.2.1. Sampling plan (group B inspection).

A sample size shall be randomly selected as specified in Table 3. If one or more defects are found, then the lot shall be rescreened and defects removed. A new sample as specified in Table 3 shall then be randomly selected. If one or more defects are found in the second sample, then the lot shall not be supplied to this specification.

Lot size	Sample size
1 to 13	100 percent
14 to 150	13 units
151 to 280	20 units
281 to 500	29 units
501 to 1,200	34 units
1,201 to 3,200	42 units

TABLE 3: Sampling plan.4.4.1.2.2. Lot definition (group B inspection).

The production lot, as defined in MIL-STD-790, is further specified as follows:

- a. A production lot consists of all connectors covered under one military specification or standard, manufactured from the specified raw materials, processed under the same specification and procedures, produced by the same type of equipment, and submitted for inspection at one time. Each production lot of assembled connectors shall be a group identified by a common manufacturing record through all significant assembly operations.
- b. Common manufacturing records and traceability shall begin with machining and molding and shall include connector assembly. Traceability of connectors to specific physical/chemical test reports of incoming raw material is not required.

4.4.1.2.3. Disposition of sample units.

Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.4.2. Periodic inspection.

Periodic inspection shall consist of group C inspection. Except where the results of this inspection show noncompliance with the applicable requirements (see 4.4.2.1.4), delivery of products which have passed groups A and B inspections shall not be delayed pending the results of this periodic inspection.

4.4.2.1. Group C inspection (periodic tests).

Periodic tests shall be performed on a 24-month basis. Samples submitted to periodic tests shall have passed groups A and B inspections.

4.4.2.1.1. Twenty-four month periodic tests.

Every 24 months, mated connector pair sample units shall be selected, configured and tested as specified below (Note: At the discretion of the qualifying activity, an alternate test sequence may be performed):

- a. Group 6, test sample quantity and configuration: Two mating plugs and receptacles shall be provided.

4.4.2.1.2. Failures.

If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed and corrective action shall be taken in accordance with 4.4.2.1.4.

4.4.2.1.3. Disposition of sample units.

Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.4.2.1.4. Noncompliance.

If a sample fails to pass group C inspection, the manufacturer shall take corrective action on the materials or processes or both, as warranted, and on all units of product which are to be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.4.3. Failures of groups A, B or C.

Lots which fail groups A, B or C and also either fail retesting or are not submitted for retesting, shall not be shipped

4.5. Qualification inspection.

Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.6) on sample units produced with equipment and procedures normally used in production. Qualification approval may be granted upon successful completion of the inspections and tests of 4.5.3 conducted on samples of 4.5.1; such approval to include classes, finishes, shell configurations, and insert arrangements other than those actually tested, providing that the capability to manufacture these parts is demonstrated by providing exhibit samples or equivalent listing. Unless otherwise specified, qualification of receptacles must be demonstrated with all compatible plugs, and qualification of plugs must be demonstrated with all compatible receptacles. Qualification of a new plating material or other specified characteristic on a connector also qualifies the same characteristic by similarity of a product with lower performance requirements, such as protective caps and covers or dummy stowage receptacles. However, qualification of a specified characteristic on a cap, cover or dummy stowage receptacle shall not qualify a connector with the same characteristic.

4.5.1. Sample size.

The following test samples, for each assembly manufactured by normal production methods, shall be provided. Tested and untested connectors, accessories and materials may be required at no cost to the Government for qualification verification by the responsible qualification activity (see 6.6).

a. Sample sizes:

Group Number	Sample Size
Group 1	1 per cable assembly type
Group 2	10 mated pairs
Group 3	1 per cable assembly type
Group 4	1 per cable assembly type
Group 5	1 per cable assembly type
Group 6	1 per cable assembly type

TABLE 4: SAMPLE GROUPS

- b. For groups 1 and 3, each verification shall be performed with each assembly type (see 3.4) and each unique combination of components (see 3.5).
- c. For groups 1 and 3, verification of mated pairs shall be performed with each plug type (see 3.5.1) mated to each receptacle type (see 3.5.2).
- d. For group 2, ten mated pairs of each plug type (see 3.5.1) mated to each receptacle type (see 3.5.2).
- e. For group 4, verification shall be performed with each complete cable assembly of each type.
- f. For group 5, verification shall be performed with each complete cable assembly of each type.
- g. For group 6, each verification shall be performed with each assembly type (see 3.4) and each unique combination of components (see 3.5).

4.5.1.1. Sample size (limited qualification).

Manufacturers that do not intend to qualify all connector types shall verify candidate connector and cables with all possible qualified mating cables and connectors. Future qualification of additional cables and connectors will require qualification as determined by the qualifying activity.

4.5.2. Preparation of samples.

Except as otherwise specified, samples shall be provided in their as-built configuration.

4.5.3. Qualification tests.

- a. Samples provided under 4.5.1a and 4.5.1b shall be divided into two similar groups. One group shall be subjected to the tests of group 1, Table 1, in the sequence specified. The other group shall be subjected to the tests of group 3, Table 1, in the sequence specified.
- b. Samples provided under 4.5.1c shall be subjected to the test of group 2, Table 1, in the sequence specified.
- c. Samples provided under 4.5.1d shall be subjected to the tests of group 4, Table 1, in the sequence specified.
- d. Samples provided under 4.5.1e shall be subjected to the tests of group 5, Table 1, in the sequence specified.
- e. Samples provided under 4.5.1f shall be subjected to the tests of group 6, Table 1, in the sequence specified.

4.5.3.1. Failures.

One or more failures, as defined in their respective requirements paragraphs, shall be cause for refusal to grant qualification.

4.5.4. Retention of qualification.

To retain qualification, the qualified manufacturer shall periodically forward reports to the Qualifying Activity (see 6.6) as follows:

4.5.4.1. At 24-month intervals, unless otherwise specified by the Qualifying Activity, the qualified manufacturer shall forward a report with a summary of the results of groups A and B testing (see 4.4). The results of tests for all reworked lots shall be identified and accounted for.

4.5.4.2. b. At 24-month intervals, unless otherwise specified, the qualified manufacturer shall submit requalification, group C test reports (see 4.4.2.1), including the number and mode of failures. The summary shall include results of all qualification verification inspection testing performed during that period.

4.5.4.3. Reports shall include certification that the qualified source continues to produce qualified product under the same conditions as originally qualified (i.e., same processes, materials, design, construction, etc., including manufacturing locations), unless such changes have been authorized by the Qualifying Activity.

Note: The qualified source shall immediately notify the Qualifying Activity at any time during the reporting period when inspection data indicates failure of the qualified product to meet the requirements of the specification. If the summary of test results indicates nonconformance with the specification requirements, and corrective action acceptable to the Qualifying Activity has not been taken, then action may be taken to remove the failing product from the QPL. In addition, failure to submit the report within thirty days after the end of each reporting period may result in loss of qualification for that product. In the event that no production occurred during the reporting period, the qualified manufacturer shall submit a report certifying that the company still has the capability and facilities required to produce the qualified product. If there has been no production during two consecutive reporting periods, the Qualifying Activity, at their discretion, may require the manufacturer submit a representative product of each type, class, etc., to retention testing in accordance with qualification inspection requirements.

Inspection	Requirement paragraph	Test paragraph
<u>Group 1</u>		

<u>(environmental)</u>		
Visual and mechanical examination.	3.1, 3.3, 3.4, 3.10, 3.35 and 3.36	0
Basic Operational Test (BOT).	3.9	4.6
Temperature shock.	3.11	4.11
High temperature exposure.	3.27	4.27
Low temperature exposure.	3.28	4.28
Low Pressure (Altitude).	3.19	4.19
Vibration.	3.20	4.20
Humidity.	3.22	4.22
Ozone exposure.	3.24	4.24
High temperature exposure.	3.27	4.27
Ice resistance.	3.29	4.29
Dust or fine sand.	3.31	4.31
Rain.	3.30	4.30
Hydrolytic stability.	3.32	4.32
Freezing rain.	3.33	4.33
Salt Fog.	3.34	4.34
Immersion.	3.25	4.25
Coupling and uncoupling force.	3.13	4.13
Insulation resistance (IR).	3.15	4.15
Dielectric withstanding voltage (DWV).	3.16	4.16
Basic Operational Test (BOT).	3.9	4.6
Post test examination.	3.35 and 3.36	4.35
<u>Group 2</u> <u>(durability)</u>		
Visual and mechanical examination.	3.1, 3.3, 3.4, 3.10, 3.35 and 3.36	0
Basic Operational Test (BOT).	3.9	4.6
Temperature shock.	3.11	4.11
Immersion.	3.25	4.25
Mating / Unmating cycles.	3.14	4.14
Coupling and uncoupling force.	3.13	4.13
Connector and cable flexing.	3.18	4.18
Basic Operational Test (BOT).	3.9	4.6
Post test examination.	3.35 and 3.36	4.35

<u>Group 3</u> <u>(Water Jets)</u>		
Visual and mechanical examination.	3.1, 3.3, 3.4, 3.10, 3.35 and 3.36	0
Basic Operational Test (BOT).	3.9	4.6
Water Jets (IP X6).	3.17	4.17
Basic Operational Test (BOT).	3.9	4.6
Post test examination.	3.35 and 3.36	4.35
<u>Group 4</u> <u>(EMI)</u>		
Visual and mechanical examination.	3.1, 3.3, 3.4, 3.10, 3.35 and 3.36	0
Basic Operational Test (BOT).	3.9	4.6
EMI/EMC.	3.23	4.23
Post test examination.	3.35 and 3.36	4.35
<u>Group 5</u> <u>(USB compliance)</u>		
Visual and mechanical examination.	3.1, 3.3, 3.4, 3.10, 3.35 and 3.36	0
USB Type-C® cable & connector.	3.9	4.9
Post test examination.	3.35 and 3.36	4.35
<u>Group 6</u> <u>(24-month periodic inspection)</u>		
Visual and mechanical examination.	3.1, 3.3, 3.4, 3.10, 3.35 and 3.36	0
Basic Operational Test (BOT).	3.9	4.6
Temperature shock.	3.11	4.11
High temperature exposure.	3.27	4.27
Low temperature exposure.	3.28	4.28
Coupling and uncoupling force.	3.13	4.13
Mating / Unmating cycles.	3.14	4.14
Insulation resistance (IR).	3.15	4.15
Dielectric withstanding voltage (DWV).	3.16	4.16
Water Jets (IP X6).	3.17	4.17
Connector and cable flexing.	3.18	4.18

Low Pressure (Altitude).	3.19	4.19
Vibration.	3.20	4.20
Humidity.	3.22	4.22
Ozone exposure.	3.24	4.24
Immersion.	3.25	4.25
Ice resistance.	3.29	4.29
Hydrolytic stability.	3.32	4.32
Freezing rain.	3.33	4.33
Basic Operational Test (BOT).	3.9	4.6
Post test examination.	3.35 and 3.36	4.35

TABLE 5: Qualification and group C periodic tests.

Visual and mechanical examination.

4.5.5. Connectors, accessories and piece parts.

The connectors, accessories and piece parts covered by this specification shall be visually and mechanically examined to ensure conformance with the specification and the applicable military standards (see 3.1, 3.3, 3.4, 3.10, 3.35 and 3.36). The contractor may use in-process controls to satisfy these requirements.

4.5.6. Contacts.

The contacts shall be visually examined for mechanical defects (see 3.1, 3.3, 3.4, and 3.36). The contacts shall be inspected through a device having approximately three power (3X) magnification. The contractor may use in-process controls to satisfy these requirements.

4.5.7. Test methods.

The following identified tests and methods assure connector integrity within typical operating conditions and applications. Alternate commercial industry standard test methods are allowed when determined to provide equivalent results; however when an alternate method is proposed, the manufacturer shall submit details of the alternate method to the qualifying activity for their review and approval prior to the performance of the test. The test methods described herein are proven methods and shall be the referee methods in the event of a dispute.

4.6. Basic Operational Test (BOT).

The following USB electrical compliance tests shall be used to demonstrate basic functionality:

<u>Category</u>	<u>Test Name</u>
Power & Safety	IR Drop (Voltage Drop)
Power & Safety	Dielectric Withstanding Voltage (Hipot)
Power & Safety	Low-Level Contact Resistance (LLCR)
Configuration	CC Continuity & E-Marker Check
Signal Integrity	Insertion Loss
Signal Integrity	Eye Diagram Test

TABLE 6: BASIC OPERATIONAL TEST

4.6.1. Continuity test.

Continuity shall be checked from cable end-to-end for each contact pair. All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit. Contacts shall be monitored for any discontinuities. A detector capable of detecting all discontinuities of 1.0 microsecond or longer shall be used.

4.7. RESERVED.4.8. RESERVED.4.9. USB Type-C® cable & connector.

Cable assemblies shall demonstrate compliance with the USB Type-C® Specification. Compliance shall be demonstrated for each connector on a cable assembly. USB-IF participation is required and connector/cable assemblies shall be certified.

Certification of compliance is not required for Plug-to-Receptacle cable assemblies (3.4.2). However, all applicable compliance tests shall be performed for these assembly types.

4.10. Interchangeability.

See 3.10. Verify through demonstration all connectors having the same Part or Identifying Number (PIN) shall be completely interchangeable with each other with respect to installation and performance.

4.11. Temperature shock.

See 3.11. Verify according to MIL-STD-810 Method 503.7 Procedure I-C. 130°F (T2) to -40°F (T1); from ambient to T1 temp (not to exceed 3°C/min); stabilize; then adjust to T2 temp in no more than 1 minute, stabilize; then adjust back to T1 in no more than 1 minute, and stabilize; repeat this two times (3 cycles total). Adjust back to ambient (not to exceed 3°C/min). Following exposure, verify operation according to 4.6.

4.12. RESERVED4.13. Coupling and uncoupling force.

See 3.13. Mated connectors shall be tested in accordance with test procedure EIA-364-114, except mated pairs shall be fully coupled in no less than 5 seconds, and shall be fully uncoupled in no less than 5 seconds. For qualification testing, mated connectors shall be coupled and uncoupled; the mating force which must be applied to facilitate full coupling and uncoupling shall be measured and recorded.

4.14. Mating / Unmating cycles.

See 3.14. The wired, assembled plugs and receptacles shall be coupled and uncoupled 10,000 cycles at a rate of 500 ± 50 cycles per hour maximum. The mating and unmating shall be accomplished so that the plug and receptacle are completely separated during each cycle. Following testing, verify operation according to 4.6.

4.15. Insulation resistance (IR).4.15.1. Insulation resistance at ambient temperature.

See 3.15.1. Unmated connectors shall be tested in accordance with test procedure EIA/ECA-364-21. The following details and exceptions apply:

- a. The tolerance on the applied voltage shall be ±10 percent.
- b. Connectors shall be mated when testing after humidity, and altitude-low temperature.

4.15.2. Insulation resistance at elevated temperature.

See 3.15.2. Unmated connectors shall be tested in accordance with test procedure EIA/ECA-364-21. The following details and exceptions shall apply:

- a. The tolerance on the applied voltage shall be ± 10 percent.
- b. Applicable elevated temperature for 30 minutes:

(1) $130^{\circ}\text{C} + 5^{\circ}\text{C} / - 0^{\circ}\text{C}$;

- c. Measurements shall be made while the connectors are still in the chamber at the specified temperature.

4.16. Dielectric withstanding voltage (DWV).

See 3.16.

4.16.1. Dielectric withstanding voltage at sea level.

Wired, unmated connectors shall be tested in accordance with test procedure EIA-364-20, method A. Connectors shall be mated when testing after humidity, and altitude-low temperature. The following details and exceptions apply:

- a. The magnitude of the test voltage shall be 100 VAC (RMS).
 - a. This does not apply to pins that would result in damage to paddle card components
- b. All contacts shall be tested. The test voltage shall be applied between each wired contact, and each adjacent contact, and the shell.
- c. The test voltage shall be maintained at the specified value for 2 seconds minimum.

4.16.2. RESERVED.

4.17. Water Jets (IP X6).

See 3.17. Unmated connectors shall be tested according to IEC 60529 second characteristic numeral 6. Connectors shall be exposed for minimum 3 minutes in each direction. Connectors may be dried with compressed air following this test. Following exposure, verify operation according to 4.6.

4.18. Connector and cable flexing.

See 3.18. Mated connectors shall be tested according to IEC 60512-4-1, Test 8d. The applied load shall be 1.5 kg; flex angle $\pm 90^{\circ}$ from vertical; Cycle Count: 30,000 cycles; Flex Speed: 30 cycles per minute. No visible mechanical damage shall be present following test. The change in Low Level Contact Resistance (LLCR) shall not exceed 10 m Ω after completion of the test. Mated pairs shall maintain continuity throughout the test according to 4.6.1. Following test, verify operation according to 4.6.

4.19. Low Pressure (Altitude).

See 3.19. Connectors shall be tested according to MIL-STD-810 Method 500.6.

4.19.1. Storage/Air Transport.

Verify according to MIL-STD-810 Method 500.6 Procedure I using 40,000 ft ASL / Dwell time x 4hrs; then return to standard ambient conditions. Mated and unmated connectors shall be tested. Following exposure, verify operation according to 4.6.

4.19.2. Operation/Air Carriage.

Verify according to MIL-STD-810 Method 500.6 Procedure II using 30,000ft ASL / Dwell time x 1hr. Mated connectors only shall be tested. Verify operation according to 4.6.

4.20. Vibration.

See 3.20. Mated connectors shall be tested according to MIL-STD-810 Method 514.8 Procedure I. 25.4mm (1 In) Peak to Peak circular motion to the table at a frequency of 5Hz. One half of the connector

pair shall be secured to the vibration table, 6 inches from the mating connection. The other connector shall be secured to a fixed point, 6 inches from the mating connection. Mated pairs shall maintain continuity throughout the test according to 4.6.1. Following exposure, verify operation according to 4.6.

4.21. RESERVED.

4.22. Humidity.

See 3.22. Connectors shall be tested according to MIL-STD-810 Method 507.6 Procedure II. Mated and unmated connectors shall be tested. Duration of the exposure shall be 10 cycles (days). Operational checks shall be conducted on mated and unmated connectors during the fifth and tenth cycles as specified, according to 4.6.

4.23. EMI/EMC.

Verify cables and connectors meet the corresponding MIL-STD-461 limits criteria for each specified requirement.

4.23.1. CS118 (Conducted Susceptibility, Personnel Borne Electrostatic Discharge).

Cables and connectors shall be tested per CS118 to the values and locations shown in TABLE XI. CS118 Discharge Table. Cables and connectors shall have a low-impedance (direct connection) to the ground reference plane. Following exposure, verify operation according to 4.6.

Discharge	Location	Type
1	Backshell	Contact (± 8 kV) & Air (± 15 kV)
2	Connector (male end)	Contact (± 8 kV) & Air (± 15 kV)
3	Shroud (female end)	Contact (± 8 kV) & Air (± 15 kV)
4	Overmold	Air (± 15 kV)
5	Strain relief	Air (± 15 kV)
6	Cable junction to strain relief	Air (± 15 kV)

TABLE 7: CS118 DISCHARGE TABLE.

4.23.2. RE102 (Radiated Emissions, Electric Field).

For the 2 MHz to 18 GHz frequency range, use the limits of Figure RE102-4 of MIL-STD-461 (the curve for Army ground). The requirement shall be met for both horizontally and vertically polarized fields above 2 MHz. The test shall be performed in a shielded anechoic chamber with the antenna placed 1 meter from the boundary of the Equipment Under Test (EUT). The EUT shall consist of the Cable Assembly connected between a representative host device and a representative load device (the "support equipment"). A minimum of a 1-meter length of the cable assembly shall be exposed and arranged in a serpentine (zig-zag) pattern on a non-conductive, non-metallic table, parallel to the front edge of the test setup boundary. The connectors and backshells are considered part of the EUT. The Cable Assembly must be actively exercising its highest-bandwidth data mode (e.g., USB 3.2 Gen 2x2, 20Gbps) and maximum power delivery profile (e.g., 240W EPR) simultaneously. The host and load devices shall be located such that their emissions do not contribute to the measurement.

4.23.3. RS103 (Radiated Susceptibility, Electric Field).

The test shall be performed in accordance with MIL-STD-461G, Method RS103, over the frequency range of 2 MHz to 2.7 GHz. The cable assembly shall be tested to the levels specified for "Aircraft Internal" and "Army" environments. Field Strength shall be 200 V/m. The requirement shall be met for both horizontally and vertically polarized fields above 2 MHz. Modulation shall be 1 kHz Pulse Modulation (PM), 50% duty

cycle. Dwell Time shall be a minimum of 3 seconds at each test frequency. The Equipment Under Test (EUT) shall consist of the Cable Assembly connected between a representative host device and a representative load device (the "support equipment"). A minimum of a 1-meter length of the cable assembly shall be exposed to the electromagnetic field, arranged parallel to the transmitting antenna. The connectors and backshells are considered part of the EUT. During exposure, the Cable Assembly must be actively exercising its highest-bandwidth data mode (e.g., USB 3.2 Gen 2x2, 20Gbps) and maximum power delivery profile (e.g., 240W EPR). Continuous data integrity checks (e.g., a file transfer with CRC verification) shall be run to monitor performance. The cable assembly shall be considered susceptible if, during exposure to the specified fields, any of the following occurs:

1. Any unintended reset, disconnect, or interruption of the link.
2. Any uncommanded change in the established USB Power Delivery (PD) contract.
3. The introduction of data corruption or bit errors beyond the acceptable threshold defined by the USB3.2 Gen 2x2 protocol.
4. Any physical damage to the cable assembly or connected devices.
5. Failure to return to full, pre-test operational status upon removal of the RF field.

4.24. Ozone exposure.

See 3.24. Wired, mated connectors shall be tested in accordance with test procedure EIA-364-14.

4.25. Immersion.

Initial qualification, see 3.25. Connector samples shall be tested in accordance with MIL-STD-810 Method 512.6 Procedure I for both fresh water and salt water for applicable dive profile. The temperature of the water shall be 18 °C ±10 °C (64 °F ±18 °F). Conditioning temperature shall be 10 °C (18 °F) above the water temperature for minimum of 2 hours. Following the immersion cycles, the connectors shall be tested for coupling force in accordance with 4.13 and dielectric withstanding voltage at sea level in accordance with 4.16.1 within 3 hours. Connectors shall also be visually or electronically checked for dry connection.

4.26. Contamination by fluids

RESERVED

4.27. High temperature exposure.

See 3.27. Connectors shall be tested according to MIL-STD-810 Method 501.7.

4.27.1. High temperature storage.

Verify according to MIL-STD-810 Method 501.7 Procedure I using High temperature cycles, climatic category A1 – Hot Dry Induced (Storage and Transit) Conditions for a minimum of 7 cycles. Mated and unmated connectors shall be tested. Following exposure, verify operation according to 4.6.

4.27.2. High temperature operation.

Verify according to MIL-STD-810 Method 501.7 Procedure II using High temperature cycles, climatic category A1 – Hot Dry Ambient Air Conditions for a minimum of 3 cycles. Mated connectors only shall be tested. During the period of maximum exposure, verify operation according to 4.6.

4.28. Low temperature exposure.

See 3.28. Connectors shall be tested according to MIL-STD-810 Method 502.7.

4.28.1. Low temperature storage.

Verify according to MIL-STD-810 Method 502.7 Procedure I using a temperature of -40°F. Stabilization time shall be 2 hours minimum. Exposure duration shall be 8 hours minimum. Mated and unmated connectors shall be tested. Following exposure, verify operation according to 4.6.

4.28.2. Low temperature operation.

Verify according to MIL-STD-810 Method 502.7 Procedure II using a temperature of -40°F. Stabilization time shall be 2 hours minimum. Mated connectors only shall be tested. Following stabilization, verify operation according to 4.6.

4.29. Ice resistance.

See 3.29. Mated connectors shall be immersed in tap water for 1 minute and then placed in an ambience of - 40° C +0°C/- 5°C for 1 hour. A minimum of three such cycles shall be performed until the connector surfaces are completely iced over. Immediately after removal from the last cycle, the frozen connectors shall be uncoupled and then recoupled. The connectors shall be uncoupled and recoupled a second time with the uncoupling and coupling force measured in accordance with 4.13. The uncoupling and coupling force should not be measured on the first uncoupling and recoupling.

4.30. Rain.

See 3.30. Verify according to MIL-STD-810 Method 506.6 Procedure I. Mated and unmated connectors shall be tested. Rate shall be 1.7 mm/min (4 in/hr) @ 18 m/s (40 mph) for 1 hour in each of the cardinal directions. Mated pairs shall maintain continuity throughout the test according to 4.6.1. Unmated connectors may be dried with compressed air following exposure. For mated and unmated connectors, following exposure, verify operation according to 4.6.

4.31. Dust or fine sand.

See 3.31. Mated and unmated connectors shall be tested in accordance with MIL-STD-810 Method 510.7. Mated pairs shall maintain continuity throughout the test according to 4.6.1. Unmated connectors may be wiped with a soft cloth and/or dried with compressed air following exposure. Following exposure, verify operation according to 4.6.

4.31.1. Blowing dust.

See 3.31. Verify using Procedure I.

4.31.2. Blowing sand.

See 3.31. Verify using Procedure II.

4.32. Hydrolytic stability.

See 3.32. Mated and unmated connectors shall be tested in accordance with ASTM D570 for long-term immersion.

4.33. Freezing rain.

See 3.33. Verify according to MIL-STD-810 Method 521.4. Mated connectors only shall be tested. Nozzle int dia: 12.5mm; Delivery rate: 100 L/min +/- 5%; Duration: Per Sq Meter of surface are to be sprayed = 1 min; Minimum test duration = 3 min; Distance from nozzle to surface = 2.5 to 3 meters. Following exposure, verify operation according to 4.6.

4.34. Salt Fog.

See 3.34. Verify according to MIL-STD-810 Method 509.7 Procedure I. Mated and unmated connectors shall be tested. Mated pairs shall maintain continuity throughout the test according to 4.6.1. Unmated connectors may be wiped with a soft cloth and/or dried with compressed air following exposure. For mated and unmated connectors, following exposure, verify operation according to 4.6.

4.35. Post test examination.

The tested connectors and contacts shall be examined for compliance with 3.35 and 3.36 to determine the effects of previous testing. Any evidence of cracking, loosening of parts, carbon tracking, excess wear, or missing parts shall be recorded.

5. PACKAGING

5.1. Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1. Intended use.

- a. On body
- b. Peripheral computer use
- c. Power transfer from alternative energy sources

6.2. Application guidance and restrictions.

Reserved

6.3. Derating factors.

Reserved

6.4. Military unique statement.

This connector is military unique because it is a miniature, high density, environment resisting, high vibration, and high corrosion connector. It is capable of operating at temperatures up to 130°F. This connector is the most versatile and most preferred for military applications. A full commercial substitute is not available.

6.5. Acquisition requirements.

Acquisition documents should specify the following:

- a. Title, number, and date of the specification.
- b. Packaging requirements (see 5.1).
- c. Nomenclature by class, type, finish, style, size, and PIN (see 1 and 3.35).
- d. Specify requirements for inclusion of covers, sealing plugs, sealing boots and /or tools.

6.6. Qualification.

With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Product List QPL No. 32812 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at <https://assist.dla.mil>.

6.6.1. Copies of "Provisions Governing Qualification SD-6".

Copies of "Provisions Governing Qualification SD-6" are available on the DLA Document Services' website at <https://assist.dla.mil>.

6.7. Definitions.

Blind-mate:	A situation in which personnel wishing to mate a connector pair can neither see nor touch the connector halves being mated and must rely on some mechanism to accomplish the mating.
-------------	--

Conditioning:	A term used to describe the preconditioning of a component by exposure to a specified environment(s), or the preconditioning of a part through some other specified process(es). Preconditioning is intended to simulate environmental wear or aging prior to additional testing.
Derating:	The operation of a device at less than its rated maximum capability in order to prolong its life. Maximum capabilities that may be derated include, but are not limited to, maximum voltage rating, current rating, and maximum temperature.
Environmental:	A term used to describe environment resistant / resisting electrical connectors. Connectors are provided with means such as temperature-resistant and fluid-resistant gaskets, seals and other features for protection against moisture, gases, dust, dirt or other contaminants.
Grommet:	The back section of the insert, molded from resilient silicone or fluorosilicone rubber, which provides environmental sealing around the wire or cable.
Hermetic:	A term used to describe hermetically sealed electrical connectors. Connectors have fused vitreous glass-to-metal seals to provide gas-tight or air-tight sealing, and feature fixed, permanently installed contacts with solderable terminations.
Mating pair:	A "mating" or "mated" connector pair is a plug and receptacle with compatible mechanical features allowing interconnection.
Polarization:	Also called "keying" or "clocking", polarization is the angular position of keys (plug shells) or keyways (receptacle shells) that are machined into the connector shell for compatible mating of a plug and receptacle. Polarization is intended to prevent mis-mating or cross-mating of connectors of the same size and type.
Scoop-proof:	Scoop-proof means that because of the connector long shell design, it is impossible for the mating plug connector to inadvertently be cocked into the mating receptacle and damage the pins or electrically short the contacts.
Self-locking:	A design that incorporates an automatic, anti-decoupling, locking mechanism that prevents the coupling ring from disengaging under vibration.
Qualifying activity:	The organizational element of the Government which grants qualification for specific product in accordance with this specification.

For additional connector definitions, see ECA EIA/ECA-622, Glossary of Electrical Connector Related Terms.)

6.8. Subject term (key word) listing.

Beryllium	Copper	Metal	Receptacle
Cadmium	Firewall	Nickel	Silver
Composite	Lead	Plug	Space grade

6.9. Guidance information.

6.9.1. Environmentally preferable material.

Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification.

6.9.2. Guidance on the use of alternative parts with less hazardous or nonhazardous materials.

This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit and function requirements of their application.

6.9.3. Guidance on performance determination of connectors.

Users and suppliers of connectors should be cognizant of the fact that the lowest performing member of a connector assembly, determines the performance characteristics of the connector. Certain combinations of connector components can be selected in the PIN system of the document in such a manner as to create a connector with degraded performance.

6.9.4. Guidance on use of inactive documents.

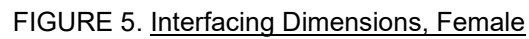
The term "Inactive for New Design" is defined by DoD 4120-24M, "Defense Standardization Program, Policies and Procedures, which is available at <http://www.dtic.mil/whs/directives/corres/pdf/412024m.pdf>. In accordance with the above-referenced policy document, an inactive document is a term applied to an existing standardization document for an item that is no longer approved for use in new designs or equipment. An inactive document can be used for procurements in support of existing designs or equipment.

6.10. Tin whisker growth.

The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM B545 (Standard Specification for Electrodeposited Coatings of Tin) (see 3.3.1.3).

6.11. Changes from previous issue.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relation to the previous issue.



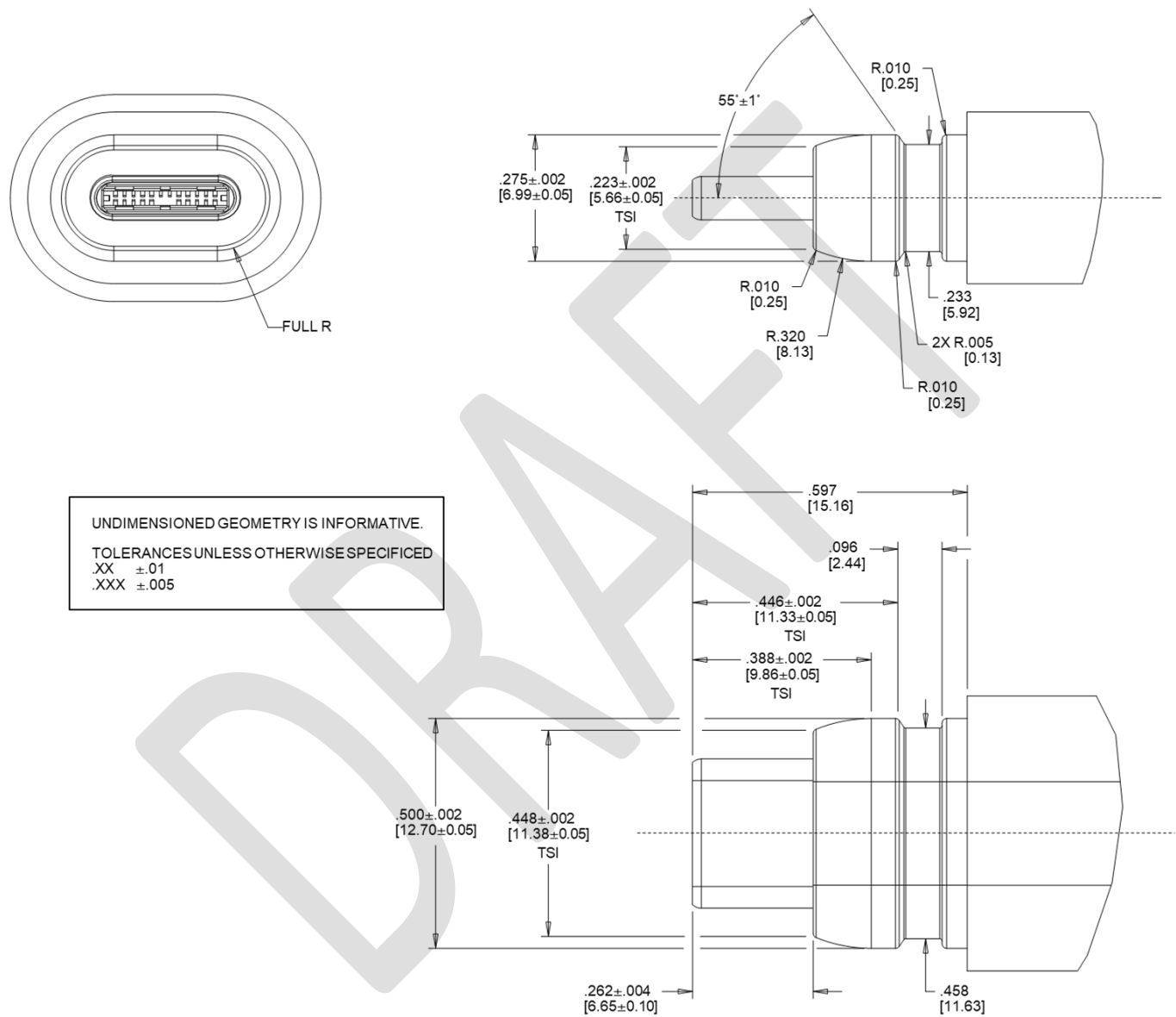


FIGURE 6. INTERFACING DIMENSIONS, MALE

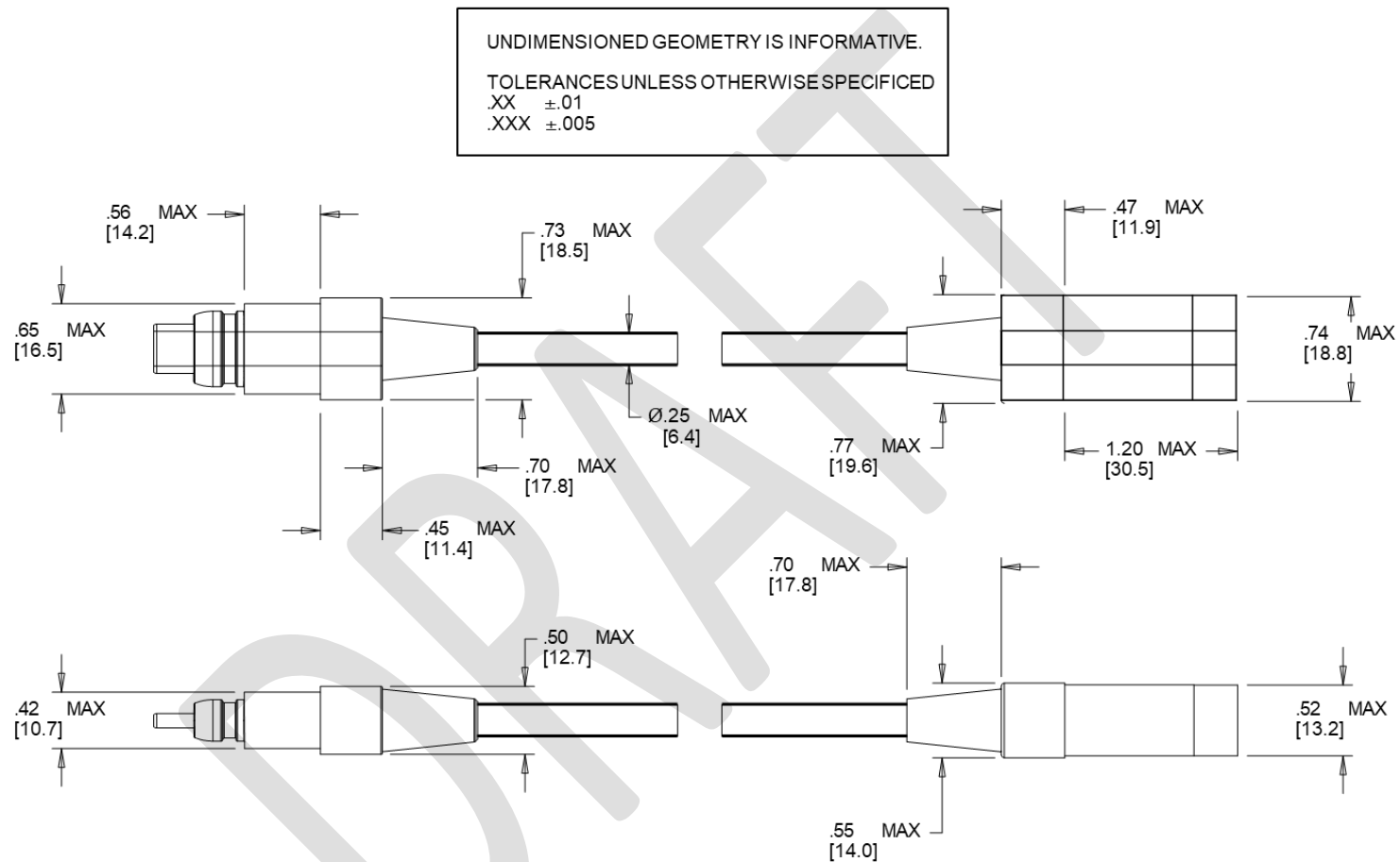
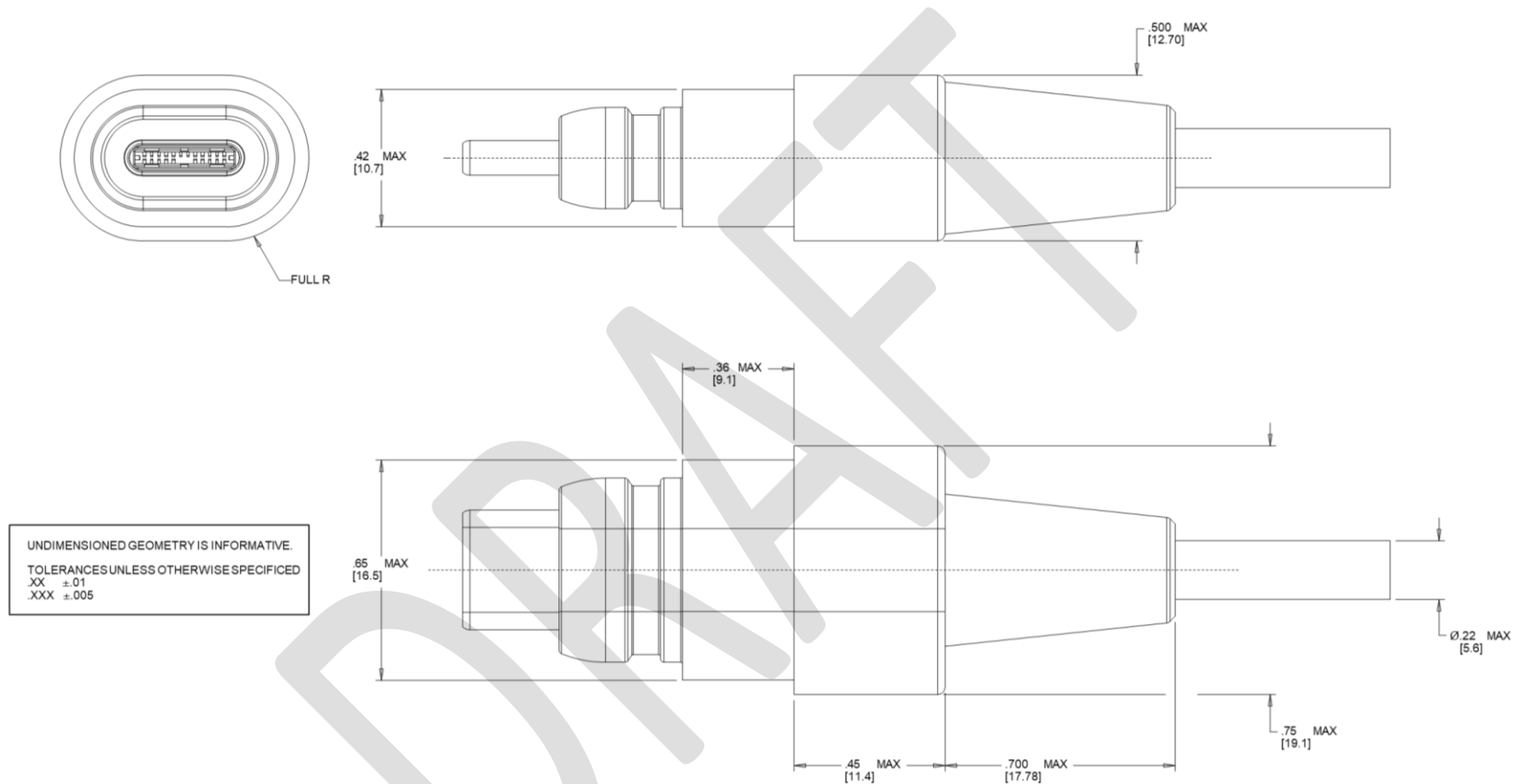


FIGURE 7. Plug, Receptacle, and Cable General Dimensions.

FIGURE 8. USB 2.0 Plug and Cable General Dimensions.

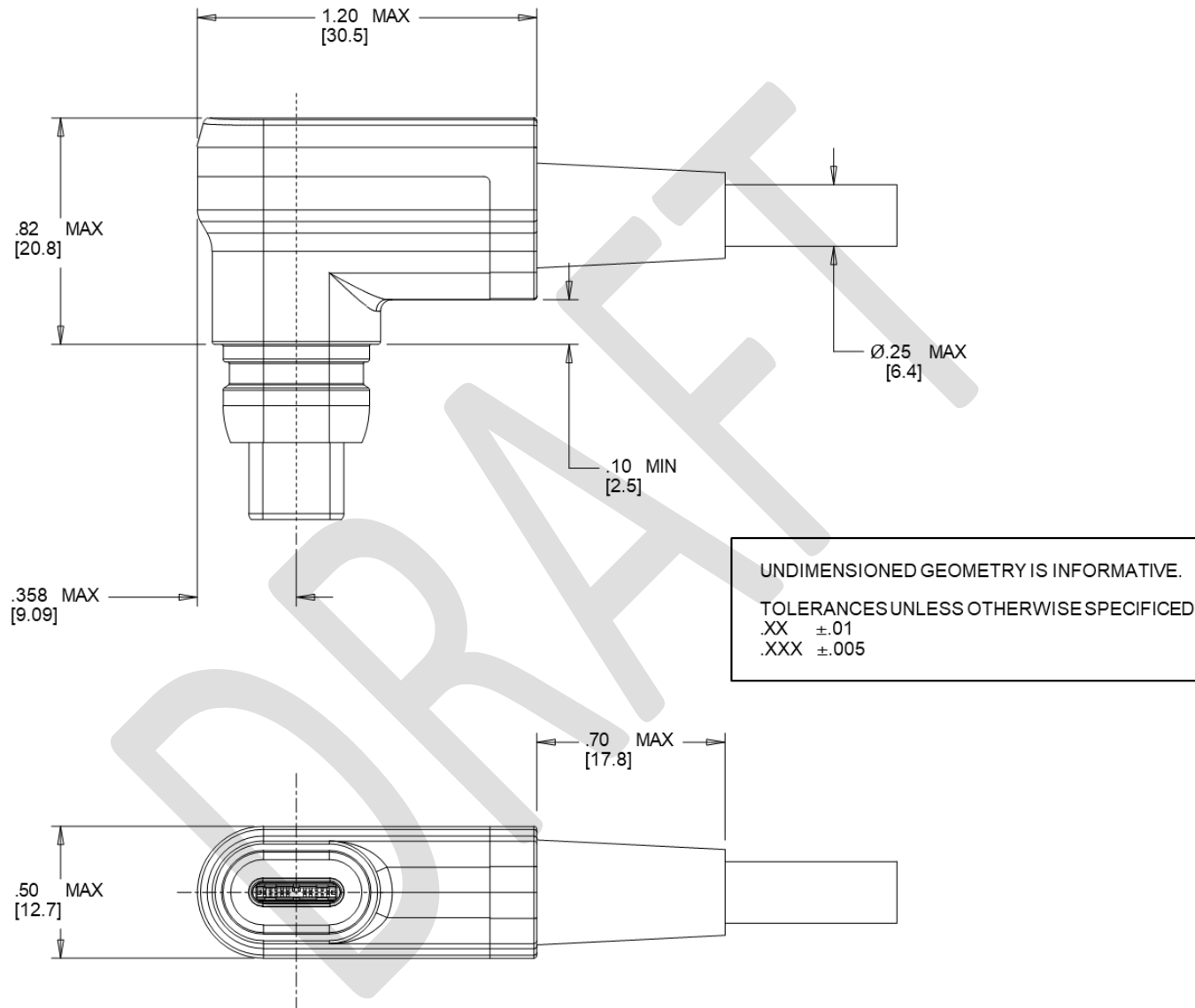


FIGURE 9. Right Angle Plug General Dimensions.

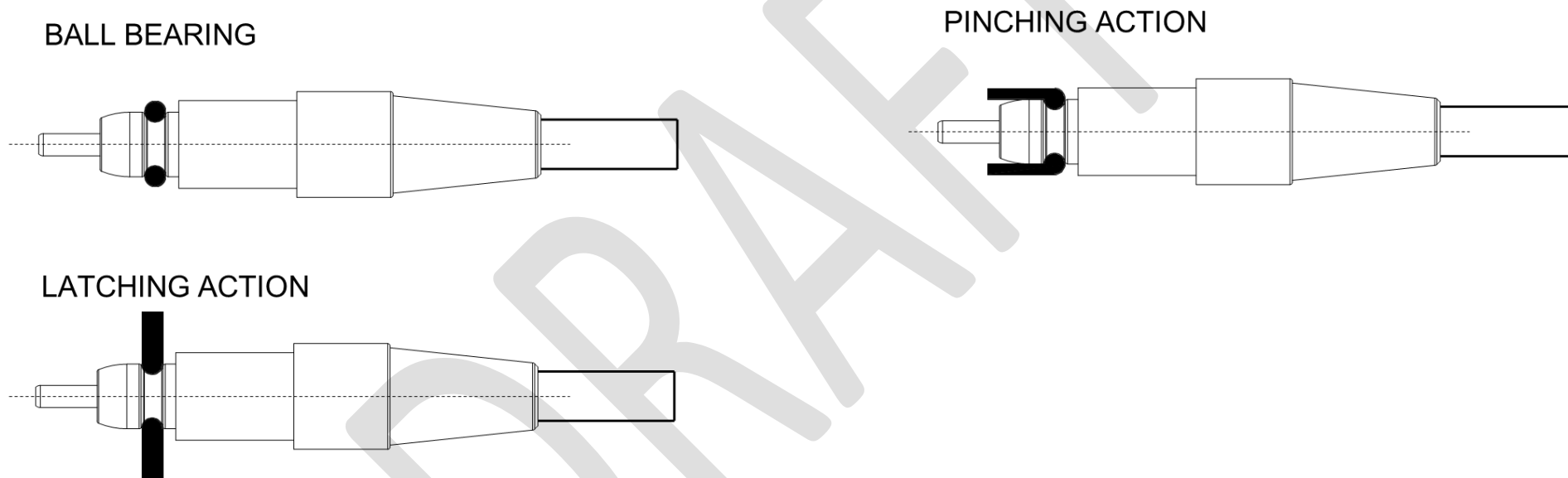


FIGURE 10. Locking Receptacle General Strategies.

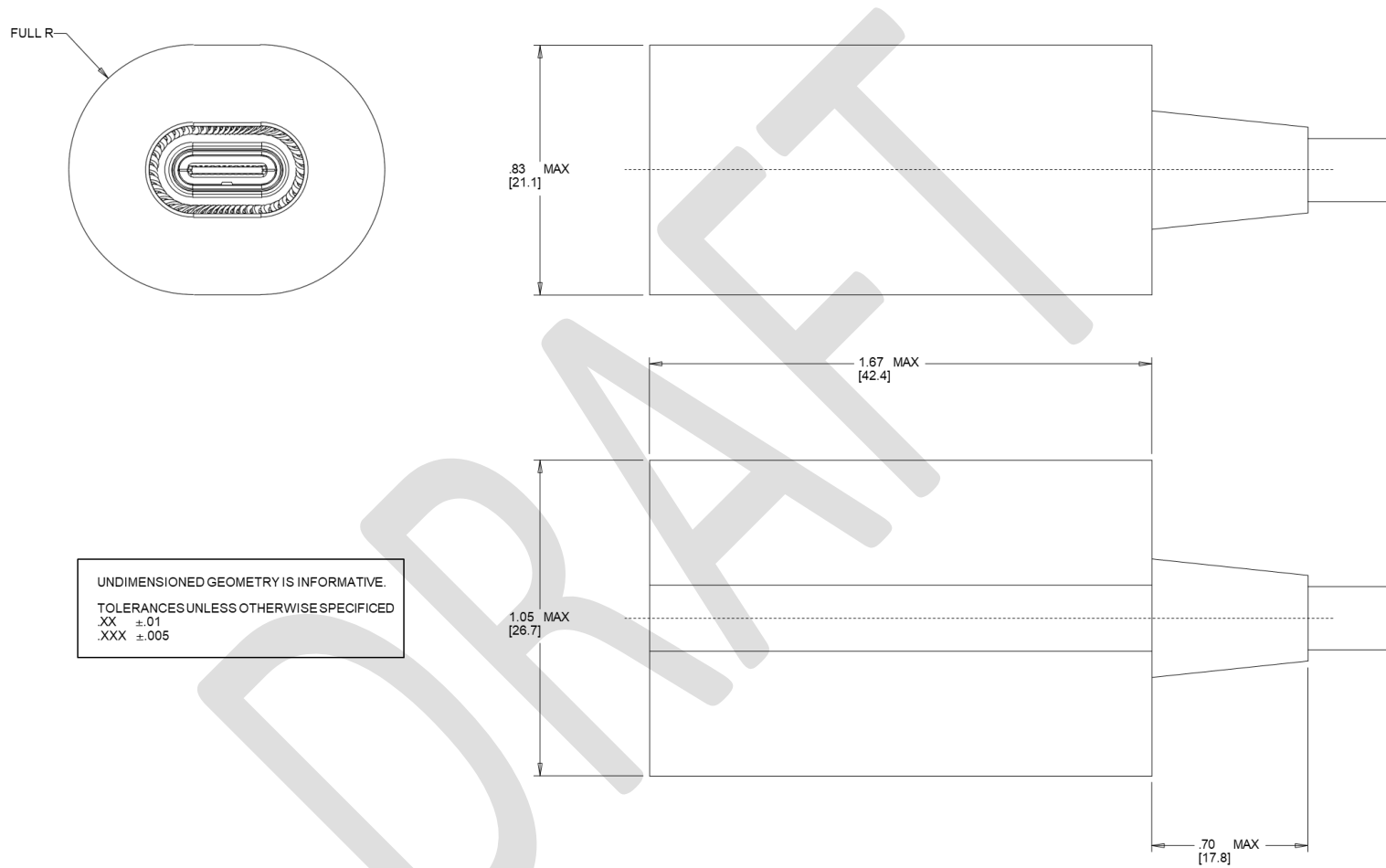


FIGURE 11 Locking receptacle max dimensions.

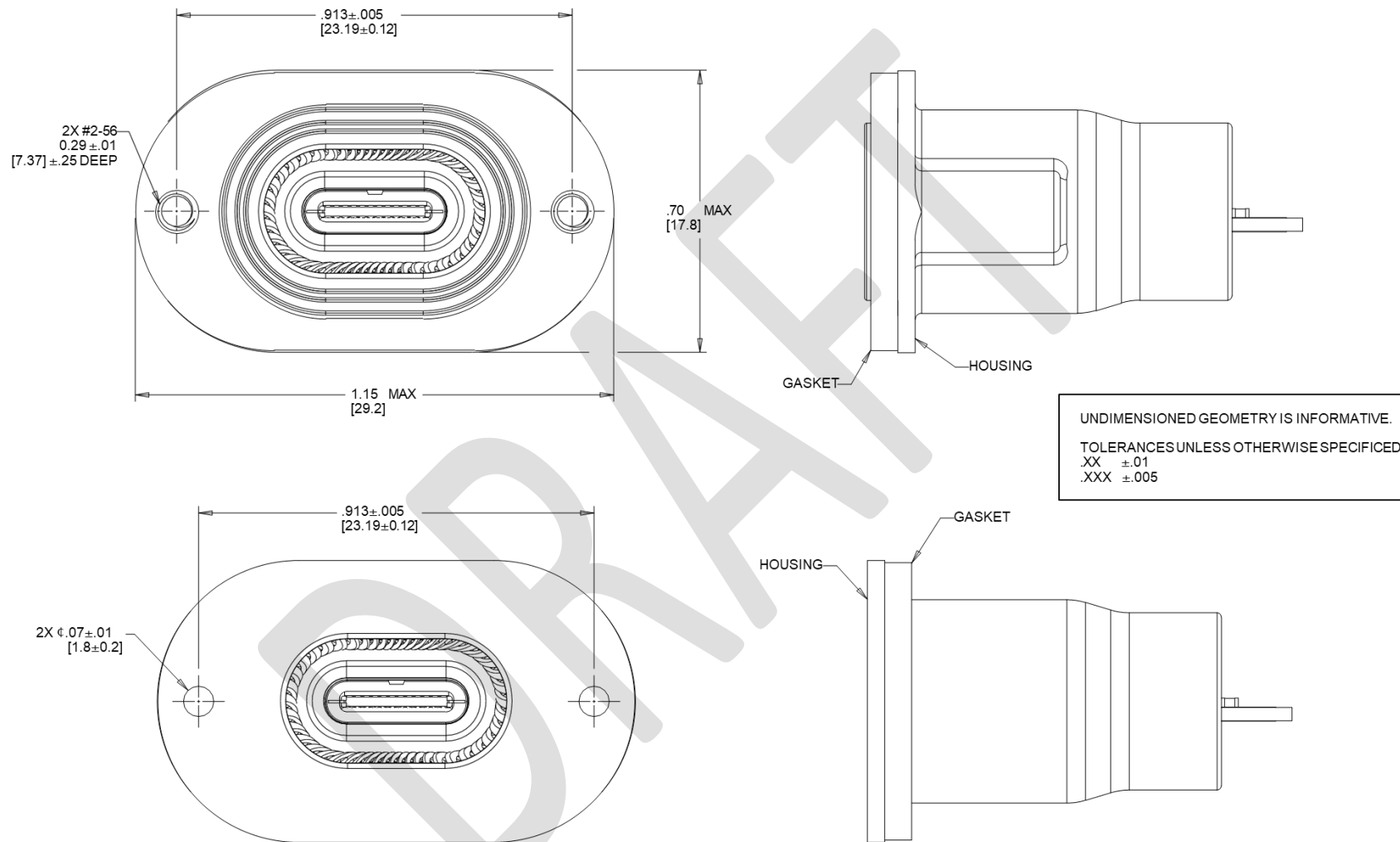


FIGURE 12. Bulkhead Receptacle General Dimensions.

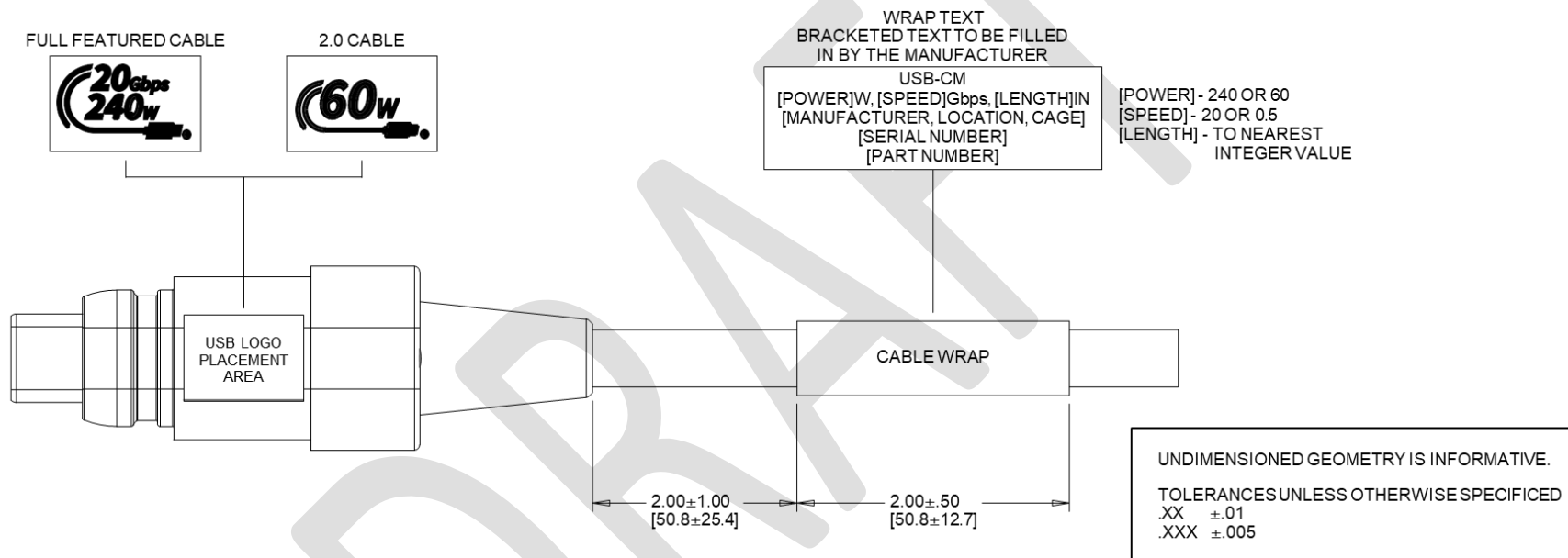


FIGURE 13. Labeling.

CONCLUDING MATERIAL

Custodians:
Army – CR, GL

Preparing activity:
Army

(Project 5998-xxxx)

Review activities:
Army – CR, GL
Other - SO

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at <https://assist.dla.mil>.